

Ivan Lopez-Mata

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About myself

Double-degree M.Sc. student in Chemistry, specializing in Physical Chemistry and Chemoinformatics through the Erasmus Mundus Chemoinformatics+ programme. Interested in computational chemistry, molecular modelling, machine learning, and data-driven molecular discovery. Experienced in electronic structure calculations, virtual screening workflows, molecular descriptors, and scientific computing environments, including Linux-based systems and workload management tools.

Education & Training

Master of Science in Chemistry (Chemoinformatics) | Université Strasbourg | 15/09/2025 - Current | Strasbourg, France

Website: <https://chimie.unistra.fr/formation/master/>

Master of Science in Chemistry (Physical Chemistry) | Università degli Studi di Milano | 15/09/2025 - Current | Milan, Italy

Website: <https://www.unimi.it/it/corsi/laurea-magistrale/scienze-chimiche>

Bachelor of Science in Chemistry | Juarez Autonomous University of Tabasco | 21/09/2020 - 05/09/2025 | Cunduacan, Tabasco, Mexico

Final grade: 9.64 | Number of credits: 288 | Website: <https://www.ujat.mx>

Work experience

Data Science Intern | SIA Software | 07/10/2024 - 24/01/2025 | Villahermosa, Tabasco, Mexico

- Design and train artificial neural networks using Python to predict disease onset based on patients' clinical data.
- Develop the front end and back end of a Hospital Information System simulator using JavaScript and Flask, integrating the trained algorithms.
- Co-author an article detailing the obtained results from the project.

Website: <https://siasw.com/index.php/en/>

Research Intern | Cluster of Excellence RESOLV - Ruhr University Bochum | 14/05/2024 - 10/08/2024 | Bochum, Germany

- Use CP2K for Density Functional Theory calculations to obtain convergence of cutoff energy and lattice parameters.
- Use CPMD for ab initio molecular dynamics to study the Pt(111)/water interface in a basic solution.
- Write reports and present them to the research group.

Department: Center for Theoretical Chemistry | Website: <https://www.theochem.ruhr-uni-bochum.de>

Research Volunteer | Chemistry Institute - National Autonomous University of Mexico | 13/09/2023 - 03/05/2024 | Merida, Yucatan, Mexico

- Develop and train Machine Learning models (SVM, KNN, Random Forest, Gradient Boosting Trees) to predict the biological activity of molecules using PubChem bioassays and molecular descriptors.
- Apply dimensionality reduction techniques (t-SNE, UMAP, PCA) to optimize data representation and evaluate model performance using key metrics such as Balanced Accuracy and F1-score.
- Co-author an article detailing the obtained results from the project.

IT Technician | Molecular Modelling Laboratory - Juarez Autonomous University of Tabasco | 13/02/2023 - 05/09/2025 | Cunduacan, Tabasco, Mexico

- Perform preventive maintenance on the computer cluster.
- Reinstall the entire cluster's operating system and compile necessary programs for computational materials science.
- Write a manual on initial computer cluster configuration for computational materials science.

Publications

Setting up computational materials science clusters

Ivan Lopez Mata (2026).

Practical and accesible guide for students, technicians and researchers looking to build a computer cluster from scratch, specifically geared towards the need of computational materials science.

Aplicación de modelos de inteligencia artificial para optimizar procesos de acoplamiento molecular en bases de datos de productos naturales

Ivan Lopez Mata, José M. Mora Fonz (supervisor) (2025).

Undergraduate graduation project completed under the case-study resolution modality, titled “Application of Artificial Intelligence Models to Optimize Molecular Docking Processes in Natural Product Databases.” The project evaluated active learning strategies to improve large-scale virtual screening using the COCONUT natural products database, combining molecular fingerprints/descriptors with machine learning models to predict docking scores and efficiently enrich high-scoring ligands while reducing computational cost.

Detection of Type 2 Diabetes Mellitus in individuals with normal fasting glucose using a multilayer perceptron neural network

2025. Rubén Alfredo Peraza-Torres, Iván López-Mata, Alejandro Medina-Santiago, Ivonne Adriana Hernández-Hernández, José Antonio Cortes-García, Jorge Antonio Orozco-Torres.

Full-paper accepted for presentation at the “2025 World Conference on Information Systems and Technologies”.

De novo design of inhibitors of DNA methyltransferase 1: A critical comparison of ligand- and structure-based approaches

2024. Diana L. Prado-Romero, Fernanda I. Saldívar-González, Iván López-Mata, Pedro A. Laurel-García, Norberto Sánchez-Cruz, José L. Medina-Franco.

Peer-reviewed journal article published at the Journal “Biomolecules”.

Social and Political Activities

President of the Student Council | 01/11/2024 - 05/09/2025 | Cunduacan, Tabasco

- Responsible for the meal and stationery support scholarship program.
- Record and analysis of student concerns to generate proposals for the university board.

President of the American Chemical Society Student Chapter | 01/11/2023 - 30/10/2024 | Cunduacan, Tabasco

- Design and maintenance of the student chapter’s website, facilitating user account creation for professors and displaying an academic staff directory.
- Collect and disseminate opportunities (scholarships, exchanges, competitions) to enhance our students’ scientific careers.
- Engage with national and international institutions, professors, and researchers to collaborate and bring diverse perspectives to our community.

Skills

Specialized Chemistry Software | CP2K | Density Functional Theory (VASP) | Quantum Chemistry software packages (Gaussian, ORCA, Multiwfn) | VMD, NAMD/Gromacs (visualization, simulation of molecular systems) | Maestro-Schrodinger | Quantum Espresso

Data Science | Python DSci/ML Libraries (Numpy, Pandas, Sklearn, Scipy) | R, R Studio, R Markdown | RDKit

Web Development | Basic knowledge of Django and Flask | Begginer JavaScript for WebDevelopment

General Tools | Exprience using Latex | Linux servers (CentOS, Kali Linux, Devuan, Ubuntu) | Queuing systems: SLURM, MOAB/Torque | Linux (commands and shell scripting)

Language Skills

Mother tongue(s): Spanish

English Understanding: Listening C1, Reading C1 | Speaking: Spoken production B2, Spoken interaction B2 | Writing: B2

Italian Understanding: Listening B2, Reading B2 | Speaking: Spoken production B1, Spoken interaction B1 | Writing: B1

French Understanding: Listening A2, Reading A2 | Speaking: Spoken production A1, Spoken interaction A1 | Writing: A1